

SEXISM IDENTIFICATION IN MEMES

A multimodal approach with text, image and physiological signals

EXIST 2026

Presented By
KLAUDIA BANASIEWICZ

PROJECT AT A GLANCE

Three sub-tasks | one dataset | text + image + physiological signals

T2.1 - Binary

Is the meme sexist?

YES / NO

T2.2 - Type

If sexist, what type?

NO / DIRECT / JUDGEMENTAL

T2.3 - Category

Multi-label: 5 sexism categories

+ NO fallback

Dataset (EXIST 2026 Memes)

3 984

training memes

1 053

test memes

EN + ES

two languages

6

annotators
per meme

**Eye + Heart
+ Brain**

sensors per meme

WHAT THE MODEL **SEES** FOR EACH MEME

Text

The caption written on the meme

- XLM-RoBERTa: a multilingual language model (handles EN + ES with one model)
- Max 128 tokens per meme

Pre-trained on 100 languages – no separate model per language.

Image

The actual picture of the meme

- CLIP (OpenAI image encoder)
- Each picture → 512 numbers that summarise 'what is in the image'

I freeze CLIP – I use its numbers but do not retrain it.

Physiological

Eye tracking + heart rate + brain signals (EEG)

- Real volunteers wore sensors while viewing each meme
- I aggregate across users – 104 numbers per meme

The model never sees raw EEG – only this 104-number summary.

MODELS I TRIED

1	Majority / Minority baseline	Always predict the most (or least) common label. A sanity-check floor.
2	Logistic regression on image + sensors	A simple linear classifier – draws a line between classes. No neural network.
3	XLM-RoBERTa-base – text only	Smaller multilingual language model, fine-tuned on the memes. 'How good is text alone?'
4	XLM-RoBERTa-large – full fine-tune	Larger version of the same model. All weights updated on the task → strong text model.
5	XLM-RoBERTa-large + LoRA + focal loss + oversampling	LoRA = tiny add-on adapters; focal loss = focus on hard examples; oversampling = copy rare examples.
6	XLM-RoBERTa-base – multi-label (5 sigmoid heads)	Five independent yes/no detectors, one per category. The model can fire on more than one.
7	Late fusion (text + image+sensors)	Two models, two probabilities. I mix them: $\text{final} = \alpha \cdot \text{text} + (1-\alpha) \cdot \text{image+sensors}$.
8	Fusion MLP – only for the cross-validation study	A small comparison network. Same model, only the inputs change → fair feature comparison.

HOW I TUNED THE TEXT MODEL FOR EACH TASK?

T2.1 - Is this sexist?

XLM-R-large - full fine-tune

- Full fine-tune: update ALL the model's weights
- 3 epochs; learning rate $1e-5$ (slow, so it does not 'forget')
- Class-weighted cross-entropy: rare-class mistakes count more
- Mixed precision (bf16) to fit in GPU memory

Why: this is the easiest task.
A strong text model is already enough.

T2.2 - What kind of sexism?

XLM-R-large + LoRA + focal loss + oversampling

- 'Judgemental' is only 14% of data → severe class imbalance
- LoRA: freeze the model, train tiny add-on adapters ($r=16$)
- Focal loss ($\gamma=2$): focus on hard examples, ignore easy ones
- Oversampling: copy rare-class memes until classes are balanced

Why: full fine-tune was unstable on the rebalanced data - LoRA is much more stable.

T2.3 - Which sexism categories?

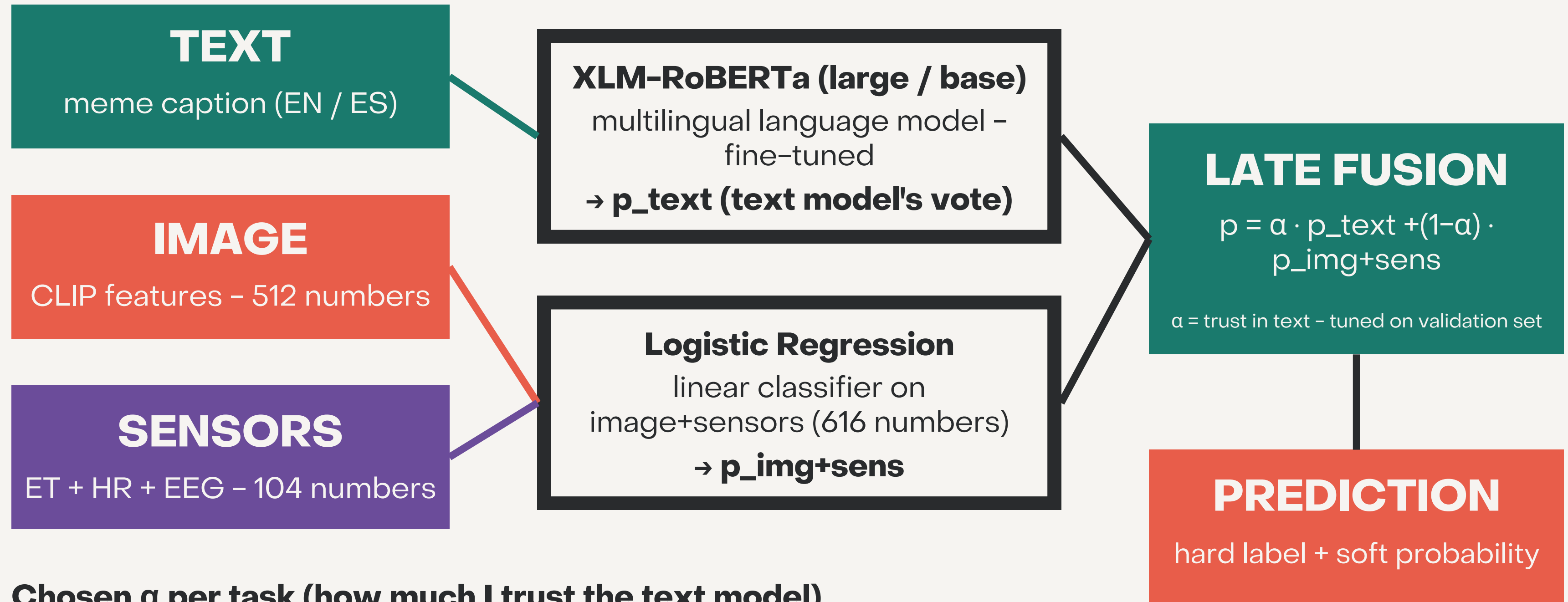
XLM-R-base, multi-label - 5 sigmoid heads

- Five independent yes/no detectors (sigmoid heads)
- Trained with BCE + pos_weight: extra weight on rare categories
- Fires above 0.5; if none fires → keep highest (top-1 fallback)
- Gated by T2.1: if T2.1 says NO → output is just [NO]

Why: a single-label head just learned 'always predict NO' → 0% on every category.

LATE FUSION – TWO MODELS THAT VOTE

Text model + image+sensor model → mix their probabilities with weight α



Chosen α per task (how much I trust the text model)

T2.1 → $\alpha = 0.95$ (95% text) · T2.2 → $\alpha = 0.75$ (75% text) · T2.3 → gated by T2.1

FINAL MODELS | SUBMITTED

One system per task – same overall idea, different recipe

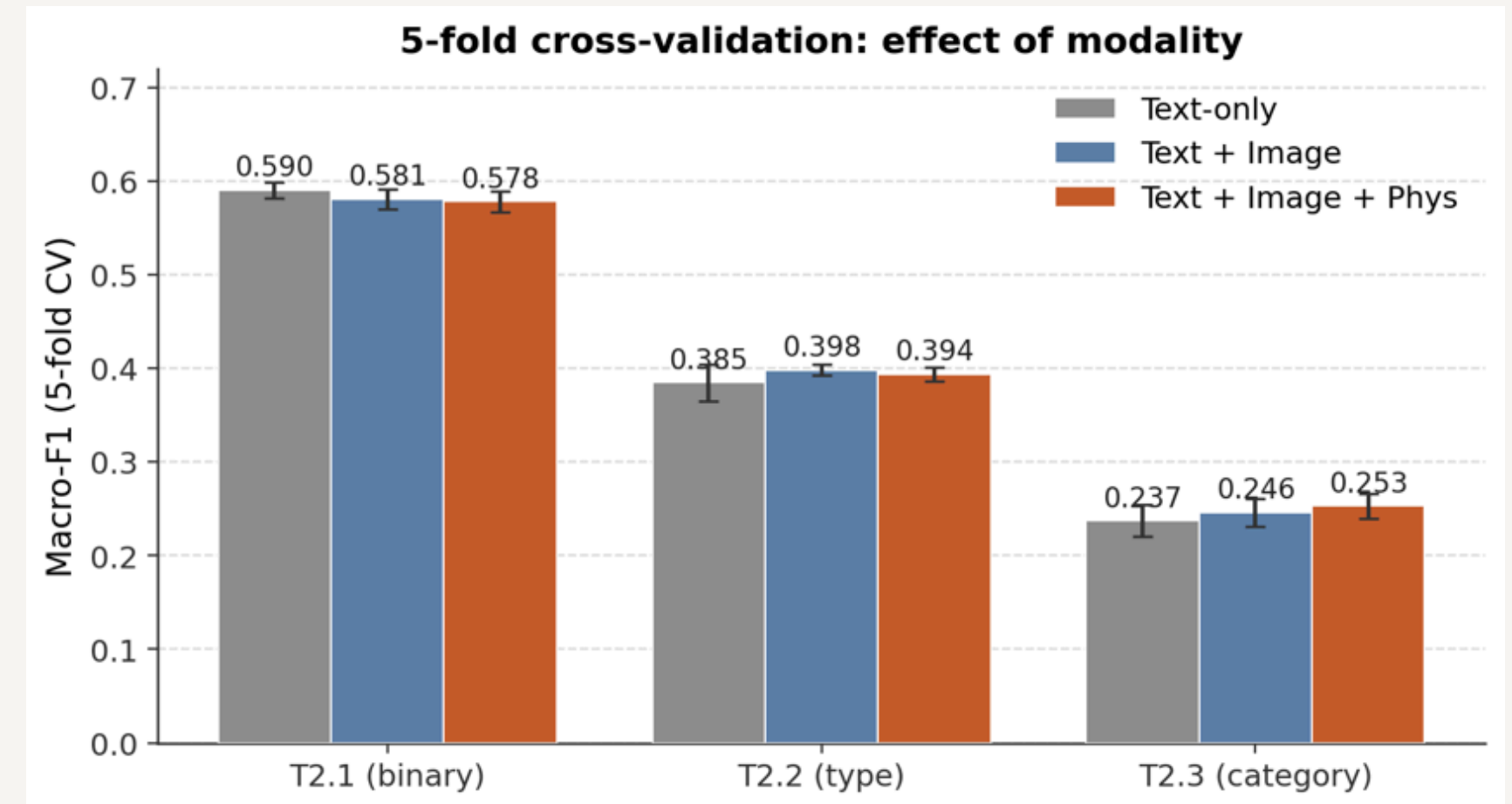
Task	Text backbone	Loss / training tricks	Fusion α
T2.1	XLM-R-large; full fine-tune (all weights)	Weighted cross-entropy; 3 epochs	$\alpha = 0.95$
T2.2	XLM-R-large + LoRA (tiny add-on adapters)	Focal loss + oversampling; 5 epochs	$\alpha = 0.75$
T2.3	XLM-R-base; 5 sigmoid heads (multi-label)	BCE + pos_weight; gated by T2.1	gated by T2.1

Why I picked these three?

- They scored the best on a held-out validation set, and they still fit on a single GPU
- All three follow the same 'two models that vote' pattern → easy to maintain and to run on the test set

5-FOLD CROSS-VALIDATION WITH VS. WITHOUT PHYSIOLOGICAL DATA

Features	T2.1	T2.2	T2.3
Text only	0.590 ± 0.009	0.385 ± 0.019	0.237 ± 0.017
Text + Image (CLIP)	0.581 ± 0.010	0.398 ± 0.006	0.246 ± 0.015
Text + Image + Phys (ET/HR/EEG)	0.578 ± 0.011	0.394 ± 0.008	0.253 ± 0.013

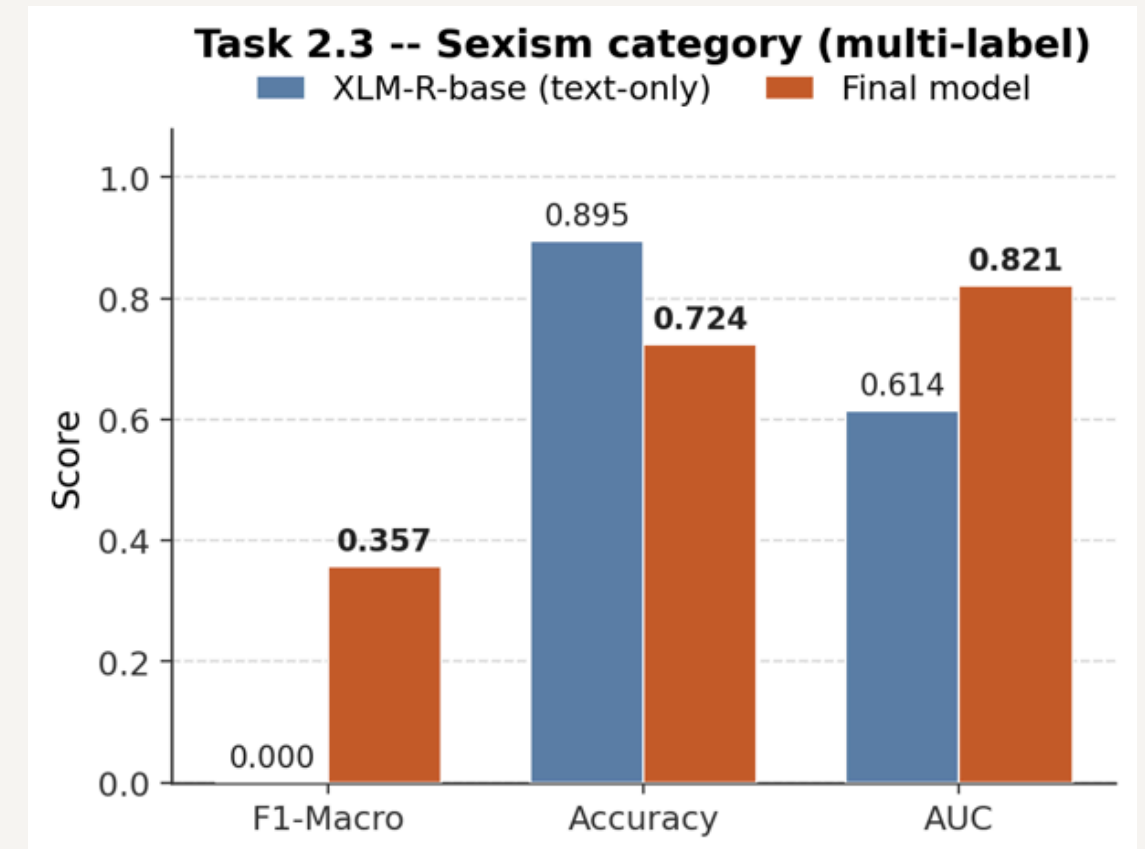
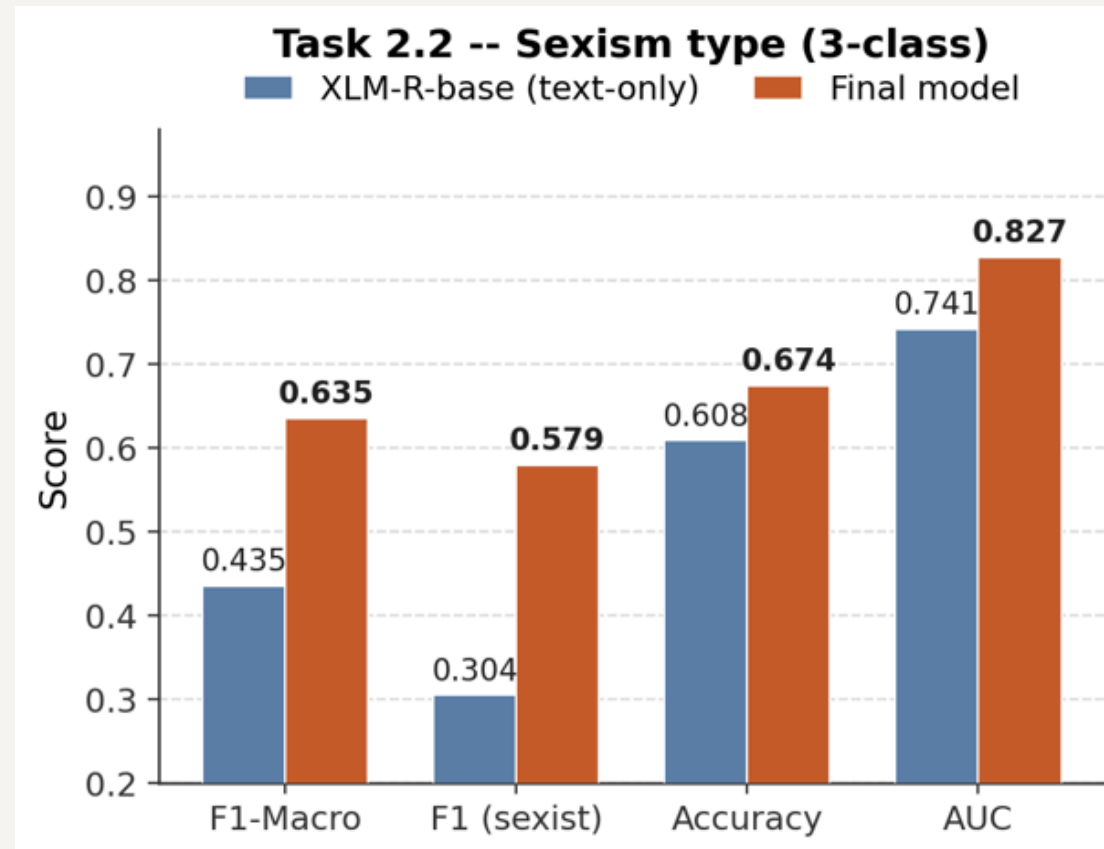
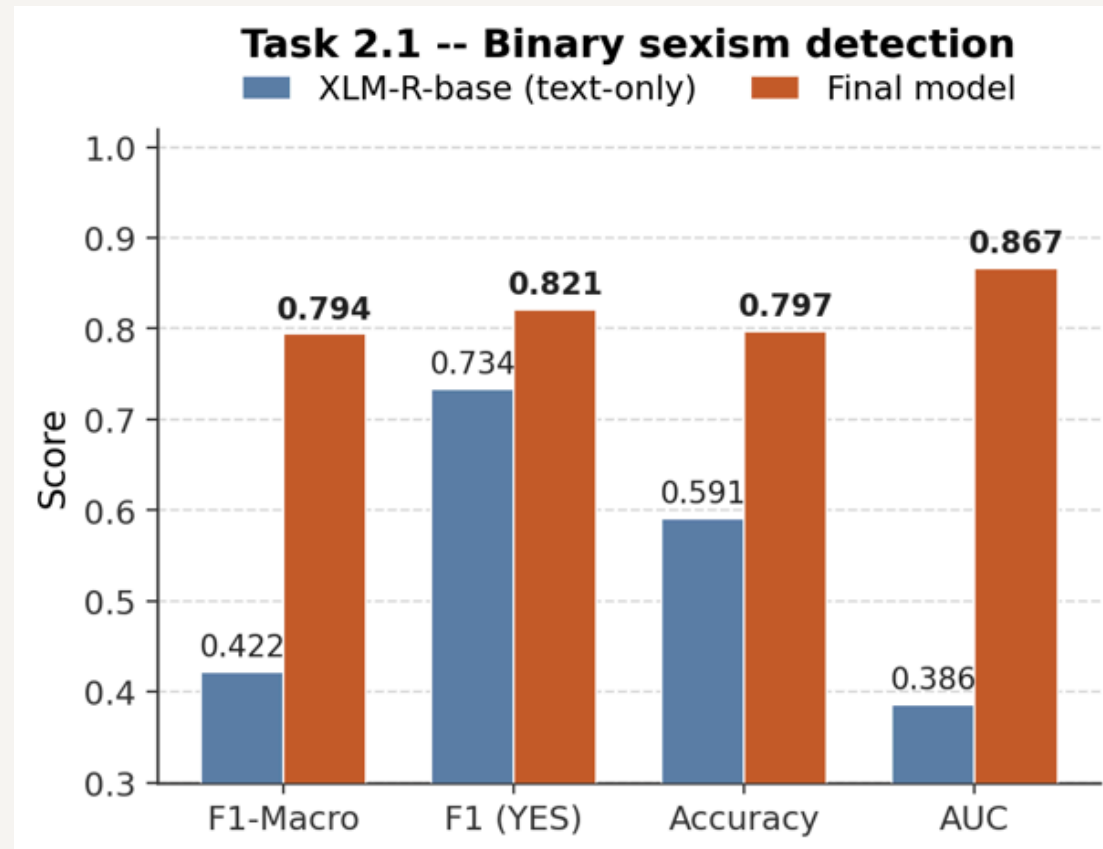


Reading the numbers (macro-F1 – averages F1 across categories, weighting each one equally; higher is better)

- **T2.1 (binary):** physiological data does NOT help – text alone already solves it. Adding sensors slightly hurts.
- **T2.2 (type):** no real change – CLIP image features already absorb the extra information.
- **T2.3 (category):** +0.016 macro-F1 from physiological data. Small, but consistent across all 5 folds → especially helps the rarest categories.
- **Take-away:** physiological signals only pay off where text saturates and class imbalance is severe (fine-grained categories).

FINAL SYSTEM VS. BASELINES

Macro-F1 and AUC on the full training set (3984 memes)



XLM-R-base text baseline → final multimodal model; F1 = score across classes, AUC = ranking quality

T2.1

F1 0.422 → 0.794

AUC 0.386 → 0.867

T2.2

F1 0.435 → 0.635

AUC 0.741 → 0.827

T2.3

F1 0.000 → 0.357

AUC 0.614 → 0.821

ZOOMING INTO T2.3

PER CATEGORY AND TEST PREDICTIONS

Categories	Final F1
Ideological inequality	0.546
Stereotyping dominance	0.392
Objectification	0.341
Sexual violence	0.285
Misogyny (non-sexual violence)	0.223
Macro average	0.357

Test-set hard predictions (1053 memes)

T2.1 580 YES 473 NO	T2.2 458 NO 346 DIRECT 249 JUDGEMENTAL
T2.3 293 OBJECTIFICATION 274 STEREOTYPING DOMINANCE 265 IDEOLOGICAL INEQUALITY 152 MISOGYNY NSV 132 SEXUAL VIOLENCE 473 PLAIN NO	

Distribution close to training frequencies → no class collapse.

TAKE-AWAYS

What worked

- Scaling XLM-R-base → XLM-R-large nearly doubled T2.1 macro-F1
- LoRA + focal loss + oversampling stabilised the rare-class T2.2 training
- Multi-label sigmoid heads + pos_weight broke the $F1 = 0$ trap on T2.3
- Late fusion (α -weighted) adds a small but real boost over text alone

What surprised me

- Physiological data (ET/HR/EEG) does NOT help binary sexism detection
- CLIP image features alone already absorb most of the multimodal gain
- Physiological data only helps the rarest T2.3 categories
- Gap between val (0.81) and CV (0.59) → XLM-R-large memorises part of the train set

If I had more time

- Replace CLIP + LogReg with a vision-language model (Qwen2-VL, LLaVA)
- Add a learnable attention over the 104 physiological features
- Train per-language models (EN / ES) on top of the shared backbone
- Confidence-weighted gating between the T2.1 and T2.3 heads

Bottom line: text is by far the strongest signal – physiological data only helps where text saturates (rare T2.3 categories).

THANK YOU

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